

A Poisson count

A Poisson model counts events in a fixed interval under a constant rate and independent increments.

What variable could count file requests during one minute?

Define the interval

A synthetic server averages 12 requests per hour under a Poisson-process model.

What is the interval for a count of requests during five minutes?

Scale the mean

A synthetic server has Poisson-process request rate 12 per hour.

What mean count applies to a five-minute interval?

Time units matter

A synthetic server has Poisson-process request rate 12 per hour.

Would $\mu = 12$ be correct for a five-minute count?

Poisson notation

A Poisson probability distribution uses mean count μ for a fixed interval. The mean is also often written λ (lambda).

$$X \sim P(\mu)$$

What does $X \sim P(2)$ mean?

Possible values

X counts requests in a fixed interval under a Poisson model.

What nonnegative integer values can X take?

Exact count probability

A synthetic Poisson model has mean 2 requests per five minutes.

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$$

What is the probability of exactly 3 requests?

No events

A synthetic Poisson model has mean 2 requests per five minutes.

What is $P(X = 0)$?

At least one event

A synthetic Poisson model has mean 2 requests per five minutes.

How can a complement calculate $P(X \geq 1)$?

Fewer than three

A synthetic Poisson model has mean 2 requests per five minutes.

Which exact probabilities add to $P(X < 3)$?

More than two

A synthetic Poisson model has mean 2 requests per five minutes.

How can a complement calculate $P(X > 2)$?

Mean and standard deviation

A Poisson count has mean μ and standard deviation $\sqrt{\mu}$.

$$\sigma = \sqrt{\mu}$$

For mean count 9, what is the standard deviation?

A spatial interval

Synthetic printing errors follow a Poisson-process model at rate 0.2 errors per page.

What mean applies to a 15-page document?

A variable rate

A server's request rate is much higher during class than overnight.

Why is one constant-rate Poisson model for both periods questionable?

Clustering

One server request automatically triggers several related requests.

Which Poisson assumption deserves scrutiny?

Binomial or Poisson?

Count successful transmissions among 100 trials, or count incoming requests during one minute.

Which count has a fixed number of trials?

A rare-event approximation

A binomial model has $n = 1,000$ and $p = 0.002$.

$$\mu = np$$

What Poisson mean would approximate its success count?

Checking a Poisson approximation

Let $X \sim B(1,000, 0.002)$. Approximate X by $Y \sim P(2)$. Use software for both probabilities.

How close are $P(X = 3)$ and $P(Y = 3)$?

Count versus waiting time

A Poisson-process model describes requests arriving in time.

How does counting requests in five minutes differ from measuring time until the next request?

Graphing a Poisson distribution

Synthetic arrivals follow $X \sim P(3)$ per hour.
Calculate $P(X = x)$ for $x = 0, 1, \dots, 10$.

How would you sketch and label the probability distribution?